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EE1030/EE2100: Matrix Theory

Lecture 19: Elementary Matrix Operations

Topics :

1. Matrix Addition and Scalar Multiplication
 2. Matrix is a Vector Space
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Addition of Matrices and Scalar Multiplication

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$. The addition of matrices A & B is defined iff $m=p$ and $n=q$ & denoted by $A+B$ is given by $(A+B)_{ij} = A_{ij} + B_{ij}$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 6 & 6 \end{bmatrix}$$

Let $A \in \mathbb{R}^{2 \times 2}$ whose column vectors are linearly independent $= [\vec{a}_1 \ \vec{a}_2]$

$B \in \mathbb{R}^{2 \times 2}$ whose column vectors are linearly independent $= [\vec{b}_1 \ \vec{b}_2]$

Let $C \in \mathbb{R}^{2 \times 2} = A+B \rightarrow$ are the column vectors of C

linearly independent? $\rightarrow$ depends on $\vec{a}_1, \vec{a}_2$ & $\vec{b}_1, \vec{b}_2$

Scalar multiplication :- If $A \in \mathbb{R}^{m \times n}$ & $\alpha \in \mathbb{R}$, then scalar mul $\alpha A \in \mathbb{R}^{m \times n}$

$$(\alpha A)_{ij} = \alpha A_{ij}$$

Two Matrices are Equal : if they are of the same size & $A_{ij} = B_{ij} \ \forall \ i, j$

Matrix as a Vector Space

Claim: Set of all matrices of size $m \times n$ whose entries come from a field F with vector addition defined as

$$(A+B)_{ij} = M_{ij} + N_{ij} \quad \forall i \in \{1, \dots, m\} \text{ \& } j \in \{1, \dots, n\}$$

and scalar mul defined as

$$(\alpha A)_{ij} = \alpha A_{ij} \quad \forall i \in \{1, \dots, m\} \text{ \& } j \in \{1, \dots, n\} \text{ \& } \alpha \in F$$

forms the vector space over field $F = (R/c)$. The vector space is denoted by $F^{m \times n}$

Transpose of a Matrix, Symmetric and Skew Symmetric Matrices

Transpose of a matrix: Let $A \in \mathbb{R}^{m \times n}$. The transpose of A , denoted by A^T is a matrix $\in \mathbb{R}^{n \times m}$ & its entries are given as

$$(A^T)_{ij} = A_{ji}$$

Symmetric Matrix: A matrix A is symmetric if $A^T = A$
 $\downarrow$
 is square

Skew Symmetric Matrix: A matrix A is skew symmetric if $A^T = -A$
 $\downarrow$
 square
 $(A^T)_{ii} = -A_{ii} \Rightarrow A_{ii} = 0$

If A is square & neither symmetric nor skew symmetric. Then, $A + A^T$ is symmetric.

$$S = A + A^T \quad \& \quad S^T = (A + A^T)^T = A^T + A = A + A^T \Rightarrow S \text{ is symmetric}$$

If A is square & neither symmetric nor skew symmetric. Then, $A - A^T$ is skew symmetric.

$$U = A - A^T \quad \& \quad U^T = A^T - A = -(A - A^T) = -U \Rightarrow U \text{ is skew symmetric}$$

Interesting Applications of Writing Matrix as a sum of Symmetric and Skew Symmetric Matrices

$\frac{1}{2}(S+U) = A \Rightarrow$ Every matrix $A \in \mathbb{R}^{n \times n}$ can be written as a sum of Symmetric & Skew Symmetric matrix

Application:

by KVL: for loop (I): $3i_1 - i_2 = 10$

for loop (II): $i_1 + 4i_2 = 0$

$$\underbrace{\begin{bmatrix} 3 & -1 \\ 1 & 4 \end{bmatrix}}_R \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

$$R = \underbrace{\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}}_S + \underbrace{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}$$

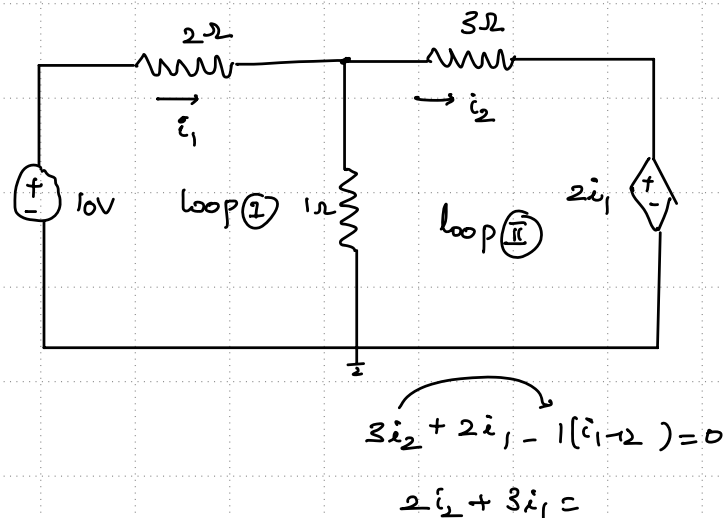
$$i_1 = 40/13$$

$$i_2 = -10/13$$

$$P_{\text{loss}} \approx 35.4 \text{ W} \rightarrow \text{①}$$

Check what the difference b/w ① & ② relate to?

Hint: Power of the dep source



$$3i_2 + 2i_1 - 1(i_1 - i_2) = 0$$

$$2i_2 + 3i_1 = 0$$

$$[i]^T [S] [i] = \begin{bmatrix} 40/13 & -10/13 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 40/13 \\ -10/13 \end{bmatrix}$$

$$= \begin{bmatrix} 40/13 & -10/13 \end{bmatrix} \begin{bmatrix} 120/13 \\ -40/13 \end{bmatrix}$$

$$= 30.7 \text{ W} \rightarrow \text{②}$$

$$[i]^T [b] = [i]^T \underbrace{[R]} [i] = \vec{e}^T [s+u] \vec{z} = \vec{e}^T [s] \vec{z} + \underbrace{\vec{e}^T u \vec{z}}_{\downarrow}$$

$$[i]^T [b] = \vec{e}^T [s] [i] \quad \text{when } \hat{0} \quad p_{\text{damp}} = (\vec{e}^T s(i))^0$$

$$\& \text{ if } (i)^T s i > 0, \Rightarrow \underline{\underline{p_{\text{damp}}?}}$$